Auteur: Seda Jusjajeva
Studierichting: Informatica
Oefeningenreeks 1C nr. 12.1

$$\frac{-1+i}{\sqrt{2}}$$

Teller:

modulus:

$$r_t = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

argument:

$$\alpha_t = \tan^{-1}\left(\frac{1}{-1}\right) = \tan^{-1}(-1) = \frac{3\pi}{4}$$

goniometrische vorm:

$$z_t = \sqrt{2}\,\mathrm{cis}\,\frac{3\pi}{4}$$

Noemer:

modulus:

$$r_n = \sqrt{\sqrt{2}^2 + 0^2} = \sqrt{2}$$

argument:

$$\alpha_n = \tan^{-1}\left(\frac{0}{\sqrt{2}}\right) = 0$$

goniometrische vorm:

$$z_n = \sqrt{2}\,\mathrm{cis}\,0$$

$$\Rightarrow \frac{\sqrt{2}\,\mathrm{cis}\,\frac{3\pi}{4}}{\sqrt{2}\,\mathrm{cis}\,0} = \frac{\sqrt{2}}{\sqrt{2}}\,\mathrm{cis}\left(\frac{3\pi}{4} - 0\right) = \mathrm{cis}\left(\frac{3\pi}{4}\right)$$

References